

The Price of Thought: A Multilingual Analysis of Reasoning, Performance, and Cost of Negotiation in Large Language Models

Anonymous ACL submission

Abstract

This document is a supplements to the general instructions for *ACL authors. It contains instructions for using the $\LaTeX$ style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

1 Introduction

Negotiation is a key aspect of human social and economic behaviour that poses significant challenges for AI systems. The problem is complex - agents must produce coherent language while performing strategic reasoning, modelling opponent preferences and goals, and managing the tension between cooperation and competition to maximize outcomes. With LLMs increasingly being deployed as autonomous agents in domains ranging from e-commerce to resource distribution, negotiation capability has become essential. Negotiation tasks therefore provide a strong test-bed for measuring agent competence, offering interactive, multi-turn scenarios with objective success measures that go beyond static evaluation and test integrated capabilities from language understanding to theory of mind.

Recent work has systematically assessed LLM negotiation performance, uncovering varied emergent behaviors and notable weaknesses. Studies consistently show that LLMs often deviate from optimal game-theoretic play, with unreliable performance across tasks (Hua et al. 2024; Bianchi et al. 2024). While larger proprietary models typically outperform smaller ones, even top-performing systems can lose to weaker opponents and fail in cooperative scenarios (Davidson et al. 2024; Kwon et al. 2024). These models can acquire complex deceptive tactics like expressing false interest in

low-value items for later concessions (Lewis et al. 2017). Yet they also exhibit cognitive biases such as anchoring effects and respond to social signals - counterintuitively, expressing desperation can improve outcomes (Bianchi et al. 2024). Despite these capabilities, serious limitations remain. LLMs show clear asymmetries, performing much worse as buyers than sellers (Xia et al. 2024). Their behavioral failures create real economic risks: budget violations, overpayment, and negotiation breakdowns (Zhu et al. 2025). Current approaches attempt to address these issues through structured, game-theoretic frameworks that constrain agent behavior and improve decision quality (Hua et al. 2024).

2 Related Work

(Gemp et al., 2024)
(Vaccaro et al., 2025)
(Bozdag et al., 2025)
(Shea and Yu, 2024)
(Sun et al., 2025)
(Jeknic et al., 2024) (clean up)
(Batra et al., 2020) (clean up)
(Bianchi et al., 2024) (negotiation arena)
(Lewis et al., 2017) (deal or no deal)
(Davidson et al., 2024)
(Hua et al., 2024)
(Kwon et al., 2024)
(Xia et al., 2024)
(Zhu et al., 2025)
(Esfandiari-Baiat and Edlund, 2024) (air balloon)

3 Dialogue Games for Evaluating Negotiation Abilities

clembench description bla bla bla (Chalamalasetti et al., 2023)

3.1 Deal or No Deal

The Deal or No Deal (DoND) game is a two-player, turn-based dialogue game designed to simulate a multi-issue bargaining scenario. Figure 1 shows a synthetic example of a successfully played episode of the game. DoND has previously been proposed in (Lewis et al., 2017) to training an AI system for negotiation. More recently, (Liao et al., 2024) used the same DoND game and proposed three different variations, two of which are used in this work. In DoND, players must communicate to reach a mutually beneficial agreement on the division of a set of items, each holding different values for each player. The game tries to evaluate negotiation skills, ability to express and understand preferences, and the ability to compromise. Aside from negotiation, the game also challenges the ability of LLMs to adhere to rules, such as using a specific syntax for final proposals.

#	A	Game Master	B
1		[A(GM) <GAME DESCRIPTION> Available items: 1 book, 2 hats, 2 balls. Player A value function: book: 0, hat: 1, ball: 4.	
2		[GM(B) <GAME DESCRIPTION> Player B value function: book: 6, hat: 1, ball: 1.	
3	[A(B)] Hi! Here are my preferences: I value balls the most, then hats, and I don't value the book at all. How about you? What do you value most?		
4		[A(B)] Thanks for sharing! I value the book most, while hats and balls are equally less important to me. Would you take both balls and one hat, and I'll take the book and the other hat? That way, we both get what we value most.	
5	[A(B)] That sounds fair! With this split, I'd get both balls and one hat, while you'd get the book and one hat, so we each get what we value most. Ready to submit this as my proposal unless you'd like to tweak anything.		
6		[A(B)] I agree with the split. I'll take the book and one hat, you take both balls and the other hat.	
7	[A(GM)] [Proposal: 1 hat, 2 balls]		
8		[GM(B)] The other player has submitted a proposal. You must now submit your proposal.	
9		[GM(B)] [Proposal: 1 book, 1 hat]	

Figure 1: Synthetic example of a DoND game episode. Note that the initial prompts given by the game master have been truncated for brevity. The game ends in a successful Pareto optimal agreement.

Game Mechanics At the start of each game, players are presented with a shared set of items, consisting of several types and quantities. In addition to the shared set of items, each player is also secretly given a different value function, mapping each item type to a non-negative integer indicating how much the player values that specific type of item. The fact that the two players value functions are different creates the central conflict in the game.

After having received the initial prompt from the game master, containing the rules of the game, the shared set of items, and the players respective set of items, players are able to exchange free-form natural language messages. These messages are transmitted directly between the players and can be in any format, provided they do not contain the special proposal syntax. This enables the players to engage in negotiation about how to divide the items, with the goal of reaching an agreement.

At any point during the conversation, or once prompted to by the game master after a turn limit is reached, the players can make a secret proposal using a specific syntax. Once one player has made a proposal, the game master instructs the other player to also make a proposal, crucially however, without revealing the content of the first proposal.

After both proposals are submitted, the game can end in either of two outcomes. Either the proposals are compatible, i.e., there are enough items to satisfy both proposals, or the proposals are conflicting. In case of compatible proposals, each player is awarded points based on total value of items they requested, according to their private value function. Should the proposals be conflicting, however, both players receive zero points.

The game master strictly enforces the rules. If a player fails to follow instructions, the game is immediately aborted. Unlike in (Liao et al., 2024) there is no re-prompting on errors, making rule adherence a critical component of successful play.

Instance Generation The game instances for the evaluation were generated programmatically, ensuring a wide variety for possible initial conditions and controlled difficulty. Each instance was generated with a maximum turn limit of 5 and each game instance features a random set of between 3 and 5 different item types. The item types for each game have been randomly sampled from a predefined list of 100 common nouns. The total number of items was set to between 5 and 8, with every item type being present at least once and the remaining capacity being randomly distributed among the item types. Finally, a random value function of non-negative integers for all item types was generated for each player.

The value functions were generated subject to the same constraints as in previous work on DoND (Liao et al., 2024; Lewis et al., 2017). The maximum value a single player can achieve by receiving all the items in the game instance is exactly 10. Ev-

ery item in the game instance has a non-zero value to at least one of the two players. And lastly, there exists at least one item that has a non-zero value to both players. These constraints are crucial for creating meaningful negotiation scenarios. The fixed maximum score keeps the value scale consistent, while the two other constraints on the value function ensure that no items are irrelevant and that it is impossible for both players to simultaneously achieve their maximum score, thus forcing a compromise.

Two of the three different game modes proposed in (Liao et al., 2024) have been evaluated. The initial prompt given to the players is identical across modes, except for the final rule which specifies the optimization objective to the players. In the **semi-competitive** mode, players are asked to maximize their own score. This is the standard negotiation setting, and it requires players to balance cooperation, to avoid zero-point conflicts, with competition, to maximize their individual share. For the **cooperative** mode of the game, players are asked to maximize the sum of both players scores. This transforms the game into a purely cooperative task, in which two rational agents should always achieve the maximum score, assuming they are able to communicate and reason about the preferences perfectly.

Evaluation Metrics In addition to the percentage of non-aborted games, the performance of players in the DoND game is also evaluated based on the scores achieved by the players. For the cooperative game mode, assessing the quality of agreement is straightforward. Since both players try to optimize for the same total score of both players, it is sufficient to use the ratio between the achieved total score to the maximum total score for that game instance, $Quality_{coop} = \frac{Score_A + Score_B}{\max(Score'_A + Score'_B)}$.

Evaluating the game in the semi-competitive setting is slightly more involved, as it requires finding a single score that jointly reflects the quality of the outcome for both players. For this, a metric based on Pareto efficiency has been used. The score is defined as one minus the normalized maximum possible Pareto improvement, $Quality_{semi} = 1 - \frac{MaximumParetoImprovement}{\max(Score'_i)}$. The maximum possible Pareto improvement is the largest amount the score of one player could be increased without decreasing the score of the other. Normalization is performed by dividing by the maximum score a single player can obtain.

3.2 Clean Up

Clean Up is a turn-based dialogue game for two players focussed on cooperative strategy development and object rearrangement. Both players are presented with identical 7×7 ASCII-like grids, with a number of randomly distributed objects in the form of capital letters placed on them. Placement is different for each player and each player can only see their own grid, cf. fig. 2. The goal is to rearrange the objects in such away that their respective positions match across both grids. Scoring is based on the Euclidean distance of the identical objects on both grids.

Object rearrangement has been a long standing challenge in the field of artificial intelligence requiring understanding of the given situation, the action space, and spatial reasoning, cf. Batra et al. (2020), Zeng et al. (2024), Khan et al. (2025).

Collaborative reconstruction games, where one player knows the desired target state and has to instruct their counterpart to recreate it, have been extensively studied (see Suglia et al. (2024) for an overview). In this game, however, the players are on par and need to both negotiate their common goal state and a strategy of reaching it (cf. Jeknic et al. (2024), which examines human-human and human-computer interaction in a similar game).

Difficulty Levels and Instances All grids have the same size (7×7), and contain obstacles in form of horizontal and vertical lines, branches, crossings, and corners cf. fig. 2. They are generated using an ASP encoding to ensure all lines are continuous. We control difficulty along two dimensions: number of empty cells and number of objects.

More empty cells means it's less likely that players try to place objects in occupied cells. Of the 49 total cells, 34 or 69.4% of cells are empty on the *easy* level, 29 or 59.2% on *medium*, and 24 or 49.0% on *hard* (cf. fig. 14 in appendix C.2 for example grids). For each level, we sample three grids, and then place 3, 5, or 7 objects on them, making for 27 experiments in total.

Game Mechanics The game is turn based, and the maximum number of rounds is fixed to $4 \times n_{obj}$ (where n_{obj} is the object count). Players can collectively accumulate $2 \times n_{obj} + 2$ penalties.

Initially, P1 (Player 1) is provided with a game description including round and penalty limit and their respective grid, following which they have to send a message to P2 (Player 2). P2 receives

#	A	Game Master	B																												
1		[P1] (GM) <GAME DESCRIPTION> 1234567 <table border="1"> <tr><td>C</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>I</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> </table> Please send a message to the other player to start the game!	C	o	o	o	o	I	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	
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2		[P1] (GM) SAY: Hi! Let's coordinate our moves. [...]																													
3		[GM] (P2) <GAME DESCRIPTION> 1234567 <table border="1"> <tr><td>o</td><td>o</td><td>C</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> <tr><td>o</td><td>o</td><td>o</td><td>o</td></tr> </table> The other player started the game by sending this message: <MESSAGE> What is your first command?	o	o	C	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	
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4		[GM] (P2) MOVE: C, (1,1)																													
5		[P1] (GM) Your message has been relayed to the other player. <GAME STATE> The other player moved an object on their grid. What is your next command?																													
6		[P1] (GM) MOVE: I, (5,6)																													
7		[GM] (P2) SAY: I think our grids are aligned now, finished?																													
8		[P1] (GM) SAY: finished!																													

Figure 2: Condensed example dialogue from Clean Up. The task is to negotiate a common goal configuration for a number of objects randomly placed on each player's grid, and move them accordingly. Finally, both players have to agree the goal is reached to end the game. //Karl-Ruilin: expand first P1 message//

the same prompt and P1's message (see fig. 12 in appendix C.1 for the full prompt). In each turn, a player can either send a message to their counterpart or move an object on their grid.

Each following turn starts with a message from the Game Master consisting of three parts: (1) feedback on the last turn, i. e., either a notification that the message has been passed on or the updated grid if they moved an object; (2) information on the game state, i. e., current and maximum penalties and rounds, and (3) either the other player's message or a notification that they moved an object.

If a player does not follow the format, tries to move an object to a non-empty space or outside the grid bounds, or tries to move an object that doesn't

exist, they receive a penalty and are re-prompted with information on the nature of their mistake. See fig. 13 in appendix C.1 for the full new turn and penalty messages.

The game ends if (1) both players agree to end it, (2) round limit is exceeded, or (3) penalty limit is exceeded.

Evaluation Metrics A game counts as failure and is not scored if the penalty limit is exceeded. The Quality Score **QS** is calculated as follows:

$$QS = DS \cdot PS \cdot 100; QS \in [0, 100]$$

Distance Score $DS \in [0, 1]$ is calculated from three components, each representing the sum of Euclidean distances for all identical objects on both grids: the Initial Distance Sum **I** at the start of the game, the Final Distance Sum **F** at the end of the game, and the Expected Distance Sum **E** that approximates the distances for randomly placed objects (see appendix C.3 for details). The Expected Distance Score **ES** and Distance Reduction Score **RS**, quantifying how close the players came to the goal of perfect alignment of all objects, are calculated as follows:

$$ES = \max \left\{ 0, 1 - \frac{F}{E} \right\}; RS = \max \left\{ 0, 1 - \frac{F}{I} \right\}$$

DS is then either 0 if final placement is worse than random, or calculated as the mean of both scores:

$$DS = \begin{cases} \frac{ES+RS}{2} & \text{if } ES > 0 \\ 0 & \text{otherwise} \end{cases}$$

Penalty Score $PS \in [0.5, 1]$ is calculated from the penalty count P normalized against max. penalties P_m as follows:

$$PS = \frac{P_m}{P - 2P_m} + 1.5$$

We chose the hyperbolic function because it is lenient for low P and harsher for P close to P_m .

3.3 Air Balloon Survival

Air Balloon Survival is a conversational negotiation game. Two players are travelling in a sinking hot air balloon and must agree to keep a set of items such that they do not exceed the balloon's maximum weight capacity. Each player is presented with the list of available items and their respective weight. Additionally, both players receive unique preference values for each item which remain hidden

from the other player. Players must then negotiate and make a deal which maximizes the utility score, i.e. the sum of preference values of items while respecting the weight constraint. During negotiation, players must respect the rules of the game which include a clear framework governing the syntactic structure of their messages and rules governing which actions they are allowed to take to advance in the game. The game ends once a deal has been made. A deal is defined as an explicit agreement by one player with a proposal their interlocutor has made. A proposal can be any subset of the set of items in the game. A simplified example of an episode can be seen in Figure 3. *Air Balloon Survival* probes the players abilities to reason and negotiate on two levels. On (1) the level of each individual player and (2) on a collective level. (1) On an individual level, each player is effectively given an instance of a version of the 0/1 Knapsack Problem. Hence, on this individual level our game probes a players ability to perform constraint-based optimization which includes carrying out arithmetic operations and performing combinatorial search over all possible deals. (2) On the collective level, we probe the players practical rationality, i.e. their ability to maximize personal utility by successfully negotiating an agreement with their counterpart who may have conflicting goals. Given that players must align by inferring their counterparts implicit preferences and intentions from observed responses, the game also connects to the broader topic of Theory of Mind (ToM) and LLMs.

Game Mechanics Both players receive initial prompts that may differ in the preference values assigned to items. They are instructed to use specified formats when making proposals and negotiating, and to follow rules such as only being allowed to accept proposals made by the other player (cf. Fig. 20 in Appendix D.1). There is no limit on the number of turns, but a game is aborted if no progress is made for eight consecutive turns. Each player may commit up to two mistakes, either by violating game rules or by violating the required syntax. In such cases, they are reprompted with a tailored message (cf. Figs. 21, 22). Messages are relayed through the game master. In some of our game instances we allow players to use a STRATEGIC REASONING tag the contents of which will not be forwarded to the other player by the game master which allows a player to ‘think out loud’ about, for example, the counterpart’s preferences or their

#	A	Game Master	B
		[A(GM)] <GAME DESCRIPTION> -Items and weights: Ball(2), Book(3), Magazine(1), Rope(4), Lamp(5) -Max Weight: 7 -Your Preferences: Ball=1, Book=2, Magazine=2, Rope=2, Lamp=1	
1		[A(GM)] PROPOSAL: Magazine, Ball, Book	
2		ARGUMENT: These items balance weight and value well for me. What do you think?	
		[GM(B)] <GAME DESCRIPTION> <Items, weights, weight limit> -Your Preferences: Ball=1, Book=1, Magazine=2, Rope=4, Lamp=5 <PLAYER A Message> [GM(B)] PROPOSAL: Rope, Ball, Magazine ARGUMENT: I value Rope highly, but not Magazine. How about we swap them?	
3		[A(GM)] <PLAYER B Message>	
4		[A(GM)] AGREE: Rope, Ball, Magazine	
5		ARGUMENT: I am indifferent w.r.t. Rope and Magazine. Fair deal.	

Figure 3: A simplified dialogue example from the *Air Balloon Survival* game. In this game, two players must negotiate and argue for their preferred set of items. To reach the goal state, one player must explicitly agree to a proposal made by the other. Messages from player A and player B are shown in red and green, respectively. Grey messages come from the game master, who provides the initial prompts and relays messages between the two players. //Tim-Kristina: I’d say add some made-up list of items and their weights from each player’s view so that we see what is there. and they are trying to negotiate for/about etc. Maybe Player 2 could accept some items but negotiate over others based on personal preferences (weights) etc. so that we see how personal preferences play a role here.//

own explicit preference values. When used with a reasoning model, the tag can be compared to the models internal reasoning traces. A game ends successfully once a player accepts a proposal made by the counterpart.

Instance Generation For instance generation, we draw either 15 or 35 items (depending on the experiment) from a randomly generated list constructed by concatenating a capital letter with a two-digit number (e.g., A42, C07). For each item i we draw a weight w_i from some discrete uniform distribution. We used $U\{1, 1000\}$ in all instances. The air balloon’s capacity is defined as a

fraction of the combined weight of all items, i.e., $W = \alpha \sum_{i=1}^n w_i$, where $\alpha \in (0, 1)$ is a scaling parameter controlling the tightness of the weight constraint. In all of our instances we set $\alpha = 0.5$, so the capacity of the air balloon equals half of the sum of the weights. For each player $p \in P$ we generate an individual valuation function assigning a preference value $v_p(i)$ to each item i . We control two dimensions of correlation when generating these valuations: (i) the correlation between weights and preferences, and (ii) the correlation between players preferences. For (i) we distinguish two cases. Firstly, there is the uncorrelated case, where preference values are drawn independently of the weights, $v_p(i) \sim U\{1, R\}$, $\forall i \in \{1, \dots, n\}$, where R denotes the maximum weight among all items. Secondly, there is the strongly correlated case, where the preference values are defined directly from the weights, $v_p(i) = w_i + q$, $\forall i \in \{1, \dots, n\}$, with $q = R/10$ controlling the correlation offset. For (ii) we distinguish three cases. In the equal case, both players share identical valuation functions, $v_1(i) = v_2(i)$ for all i , thus giving them the same goals. In the independent case, the valuations v_1 and v_2 are generated separately. In the inverse case, the rankings of the items are reversed across players, so that the highest-valued item of player 1 receives the lowest value for player 2, and so forth, giving them opposing goals. Increasing the number of items and controlling the correlation between weights and preferences relates to (1), the hardness of the knapsack problem and the reasoning required on an individual players level. The correlation between players preferences relates to (2), their ability to negotiate, take appropriate actions and reach an agreement when they have common or opposing goals.

Evaluation Metrics Each player $p \in P$ receives an individual score based on the utility of the final deal D , i.e. the sum of the preference values of items in D . This sum gets normalized by the value of the optimal solution to their instance of the 0/1 Knapsack Problem. The overall game score is defined as the harmonic mean of the two players normalized scores, further normalized by the maximum harmonic mean achievable for an instance. All scores are scaled to the range $[0, 100]$. If the total weight of the deal exceeds the capacity W , the score is set to 0. We chose the harmonic mean as our main metric as it rewards collectively balanced outcomes by favouring deals where both players

achieve high scores and penalizing those where one player benefits disproportionately. The choice for individual scores is straightforward, as they simply measure each players outcome relative to the optimal solution of their own knapsack problem. For a final deal $D \subseteq \{1, \dots, n\}$, the normalized score of player $p \in P$ is

$$f_p(D) = 100 \cdot \frac{\sum_{i \in D} v_p(i)}{\text{OPT}_p},$$

where OPT_p is the value of the optimal solution to player p 's 0/1 Knapsack Problem. The harmonic mean of the players' normalized scores is defined as

$$f_{\text{harm}}(D) = \begin{cases} \frac{2f_1(D)f_2(D)}{f_1(D)+f_2(D)} & \text{if } \sum_{i \in D} w_i \leq W, \\ 0 & \text{otherwise.} \end{cases}$$

The final game score is the normalized harmonic mean,

$$\text{score}(D) = 100 \cdot \frac{f_{\text{harm}}(D)}{\text{OPT}^*},$$

where OPT^* denotes the maximum harmonic mean achievable in the given instance. //Tim-Kristina: Game desc etc.//

(Esfandiari-Baiat and Edlund, 2024)

(Davidson et al., 2024)

4 Experimental Setup

4.1 Instances

4.2 Evaluation Metrics

Each game provides a

4.3 Evaluated Models

We have evaluated both commercial and open-weight models that have reasoning functionality. We selected *GPT-5*, *GPT-5-mini*¹, *Claude-4* from commercial models². From open-weight models, we selected *Llama3.3-70B* and its reasoning counterpart *Deepseek-R1-distilled-llama-70B*³ and *Nemotron-Nano-9B-v2*⁴. We have excluded certain reasoning models (e.g., *gpt-oss*) since it was

¹Reasoning mode for OpenAI models is controlled by setting the *reasoning_effort* parameter to *medium* and it was turned off by setting it to *minimal*

²Reasoning mode is controlled by setting the *thinking* parameter to true/false and providing an upper limit budget (we set it as 1024 tokens)

³*Llama3.3-70B* model was finetuned to make it a reasoning model by distilling from *Deepseek-R1*

⁴It is a reasoning model and reasoning can be turned off by including *no_think* in the *system message* and by default it is turned on

not possible to turn off their reasoning mode. For commercial models, we ran them via their respective API backends and we used *OpenRouter.ai* for open-weight models.

5 Results

5.1 Overall Performance

5.2 Findings

Deal or no deal:

Clean Up:

Air Balloon Survival:

5.3 Cost-Performance Trade-off

5.4 Qualitative Analysis

6 Conclusion

Limitations

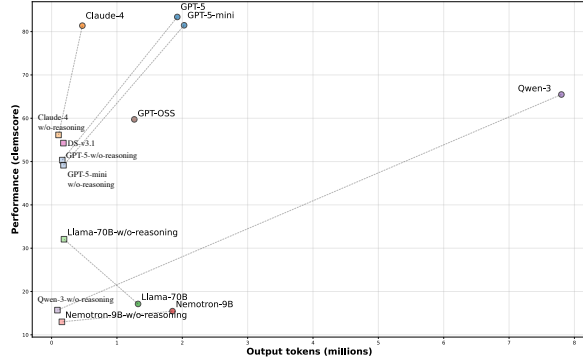
LIMITATIONS

References

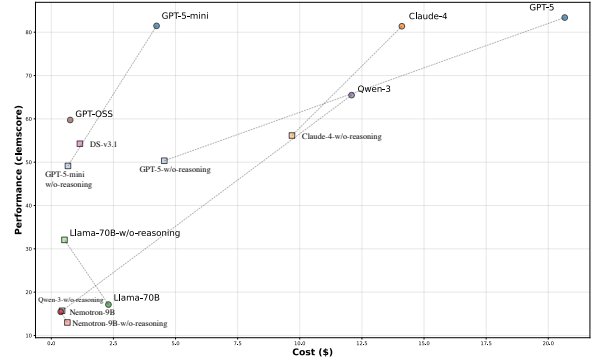
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		GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3 80B		GPT OSS	DS-v3.1
		On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
EN	DoND	87.9	32.3	75.7	23.7	94.4	90.1	24.3	43.5	18.5	4.0	57.9	22.5	50.0	59.0
	Cleanup	59.9	42.8	63.4	38.1	69.7	60.0	1.5	19.8	21.8	27.9	55.3	18.6	67.7	57.5
	Air Balloon	98.0	80.1	97.5	82.2	83.8	26.2	2.3	40.9	14.3	13.2	88.7	0	78.2	81.7
	Average	81.9	51.7	78.9	48.0	82.6	58.8	9.4	34.7	18.2	15.0	67.3	13.7	65.3	66.1
DE	DoND	86.0	19.6	82.9	24.4	80.3	77.1	21.8	53.4	0	4.5	62.1	12.8	34.9	42.4
	Cleanup	78.7	41.2	76.8	41.2	82.7	55.8	3.9	15.4	21.5	0	58.2	14.5	50.1	29.4
	Air Balloon	98.0	81.4	94.9	86.2	85.4	27.8	51.5	43.7	17.4	16.3	76.0	8.8	90.7	75.3
	Average	87.6	47.4	84.9	50.6	82.8	53.6	25.7	37.5	13.0	6.9	65.4	12.0	58.6	49.0
IT	DoND	78.0	33.8	77.1	26.8	84.4	67.8	17.3	14.3	13.9	11.2	55.6	19.4	29.0	28.9
	Cleanup	66.9	39.7	67.5	39.5	72.9	58.5	3.4	17.9	13.2	9.8	52.5	13.0	44.8	41.3
	Air Balloon	97.2	88.0	97.3	87.8	79.4	27.0	2.7	40.8	0	0	77.8	12.9	83.7	76.4
	Average	80.7	53.8	80.6	51.4	78.9	51.1	7.8	24.3	9.0	7	62.0	15.1	52.5	48.9

Table 1: *clemscore* values for three negotiation games on selected LLMs for English, German, and Italian versions. *On*: reasoning mode is turned on, *Off*: reasoning mode is turned off. The best result for each language in each row is highlighted in bold. *CL*: Claude, *LM*: Llama-3.3, *DS*: Deepseek, *Nem*: Nemotron-v2, *DoND*: Deal or No Deal



(a) Average performance (clemscore) and the number of output tokens for all evaluated models with their reasoning on/off modes for English, German and Italian.



(b) Average performance (clemscore) and the cost of for all evaluated models with their reasoning on/off modes for English, German and Italian.

Figure 4: Trade-off between performance and cost comparison across different models. Results for all languages are averaged.

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A Additional Analysis

A.1 Token Usage per Language

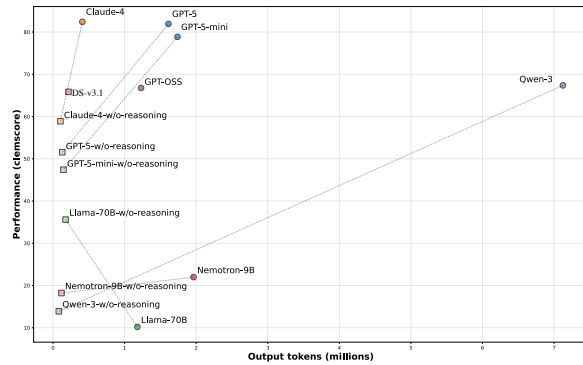


Figure 5: Performance (clemscore) and average number of output tokens for all evaluated models with their reasoning on/off modes for English.

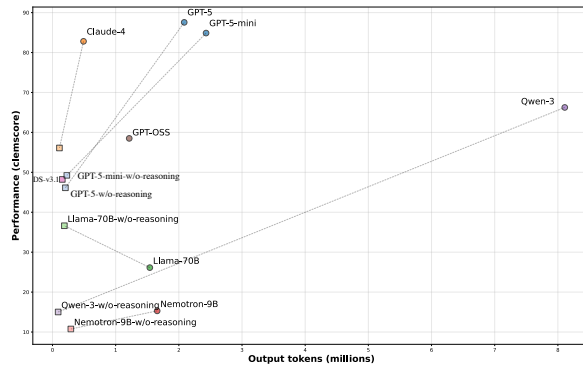


Figure 6: Performance (clemscore) and average number of output tokens for all evaluated models with their reasoning on/off modes for German.

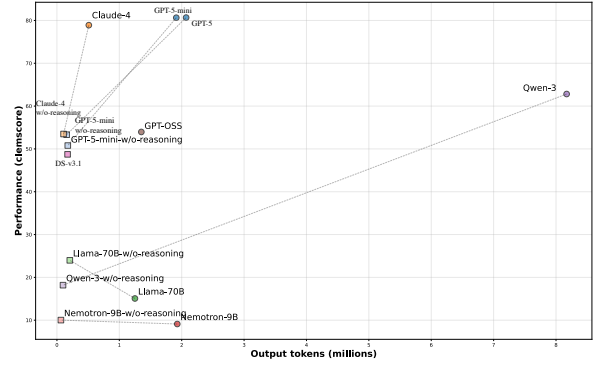


Figure 7: Performance (clemscore) and cost for all evaluated models with their reasoning on/off modes for English.

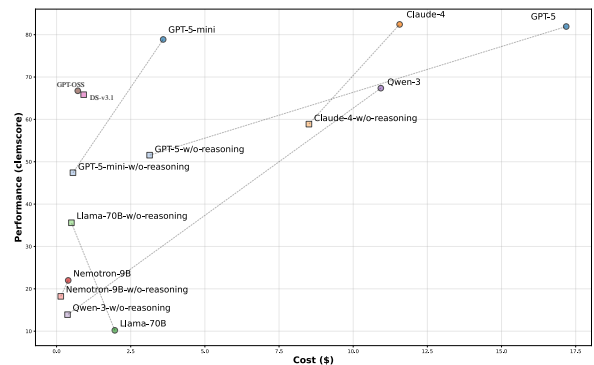


Figure 8: Performance (clemscore) and cost for all evaluated models with their reasoning on/off modes for German.

A.2 Cost per Language

B Deal or No Deal - Game Details

B.1 Prompt Templates

The prompt template for both players of the Deal or No Deal is given in Figure 11.

B.2 Experimental Setup

B.3 Overall Results

B.4 Detailed Analysis

B.5 Qualitative Samples

C Clean Up - Game Details

C.1 Prompt Templates

The prompt templates for both players of Clean Up are given in Figure 12.

Intermittent messages sent by the Game Master to the players are given in Figure 13. Following a player’s response, they are either penalized and reprompted (cf. Figure 13a, 13b), or the turn is passed to the other player (Figure 13c).

		Claude		Deepseek		GPT5		GPT5-mini		GPT-OSS	LLAMA	Nemotron		Qwen3
		Reason	No reason	r1-distilled	v3	Reason	No reason	Reason	No reason			Reason	No reason	Thinking
Optimal	Explicit maximization	0	1	0	3	11	3	11	0	1	0	0	0	5
	Simple negotiation	17	16	0	7	3	2	6	2	5	5	0	0	8
	Luck	0	0	0	0	0	0	0	0	0	0	1	0	0
	Unclear preferences	1	1	1	1	0	1	0	0	1	1	0	0	0
Suboptimal	Unclaimed items	0	0	0	0	0	0	0	0	0	1	0	0	1
	Claiming valueless item	1	2	0	0	0	2	0	1	3	4	0	0	0
	Wrong proposal	0	0	1	0	0	0	0	0	1	0	0	0	0
	Premature proposal	0	0	3	1	3	0	1	0	3	0	3	1	0
Failed	Wrong proposal	0	0	3	7	1	11	0	13	1	5	0	0	4
	Premature proposal	1	0	12	1	1	0	1	2	4	0	14	19	0
Aborted	Wrong syntax	0	0	0	0	0	1	0	1	1	4	0	0	1
	Non existant items	0	0	0	0	0	0	0	0	0	0	2	0	0
	No proposal	0	0	0	0	0	0	0	0	0	0	0	0	1

Table 2: Game outcomes for the cooperative DoND game mode.

		Claude		Deepseek		GPT5		GPT5-mini		GPT-OSS	LLAMA	Nemotron		Qwen3
		Reason	No reason	r1-distilled	v3	Reason	No reason	Reason	No reason			Reason	No reason	Thinking
Optimal	Explicit maximization	0	1	0	1	0	0	0	0	0	0	0	0	0
	Simple negotiation	18	13	1	6	14	3	0	0	2	5	0	0	6
	Luck	0	0	2	0	0	0	0	0	1	0	2	0	0
	Unclear preferences	0	0	0	1	0	0	0	0	0	0	0	0	0
Suboptimal	Unclaimed items	0	0	0	0	1	1	0	0	0	1	0	0	3
	Claiming valueless item	1	3	0	5	2	1	0	0	0	2	0	0	2
	Wrong proposal	0	0	0	0	0	0	0	0	0	0	0	0	0
	Premature proposal	0	0	1	0	2	0	0	0	0	0	2	1	0
Failed	Wrong proposal	0	3	5	6	1	12	0	0	5	2	0	0	2
	Premature proposal	1	0	7	0	0	1	0	0	2	1	14	18	6
Aborted	Wrong syntax	0	0	4	1	0	0	0	0	0	9	2	0	1
	Non existant items	0	0	0	0	0	0	0	0	0	0	1	1	0
	No proposal	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 3: Game outcomes for the semi-competitive DoND game mode.

		Claude		Deepseek		GPT5		GPT5-mini		GPT-OSS	LLAMA	Nemotron		Qwen3	Instruct
		Reason	No reason	r1-distilled	v3	Reason	No reason	Reason	No reason			Reason	No reason	Thinking	
	Tell exact values	0	3	1	13	19	17	12	14	1	0	0	0	13	17
	Tell only relative values	20	17	0	5	0	1	0	7	0	18	0	0	2	1
	Do not tell preferences	0	0	7	2	0	0	2	0	5	1	0	0	5	0
	Do not communicate at all	0	0	12	0	0	0	0	0	1	0	20	20	0	2

Table 4: Model willingness to share its value function for the cooperative DoND game mode.

		Claude		Deepseek		GPT5		GPT5-mini		GPT-OSS	LLAMA	Nemotron		Qwen3	Instruct
		Reason	No reason	r1-distilled	v3	Reason	No reason	Reason	No reason			Reason	No reason	Thinking	
	Tell exact values	0	1	0	11	0	4	0	0	2	0	0	0	6	6
	Tell only relative values	19	19	1	9	15	14	0	0	4	20	0	0	1	10
	Do not tell preferences	1	0	8	0	5	0	0	0	3	0	0	0	12	0
	Do not communicate at all	0	0	11	0	0	0	0	0	1	0	20	20	1	4

Table 5: Model willingness to share its value function for the semi-competitive DoND game mode.

		Claude		Deepseek		GPT5		GPT5-mini		GPT-OSS	LLAMA	Nemotron		Qwen3	Instruct
		Reason	No reason	r1-distilled	v3	Reason	No reason	Reason	No reason			Reason	No reason	Thinking	
	Lie about preferences	1	1	0	9	0	0	0	1	0	2	0	0	0	0
	Dont lie about preferences	19	19	20	11	19	20	19	18	20	18	20	20	20	20

Table 6: Model truthfulness about its value function for the cooperative DoND game mode.

		Claude		Deepseek		GPT5		GPT5-mini		GPT-OSS	LLAMA	Nemotron		Qwen3	Instruct
		Reason	No reason	r1-distilled	v3	Reason	No reason	Reason	No reason			Reason	No reason	Thinking	
	Lie about preferences	0	5	0	6	0	0	0	0	1	3	0	0	0	1
	Dont lie about preferences	20	15	20	14	20	18	0	0	0	17	20	20	20	0

Table 7: Model truthfulness about its value function for the semi-competitive DoND game mode.

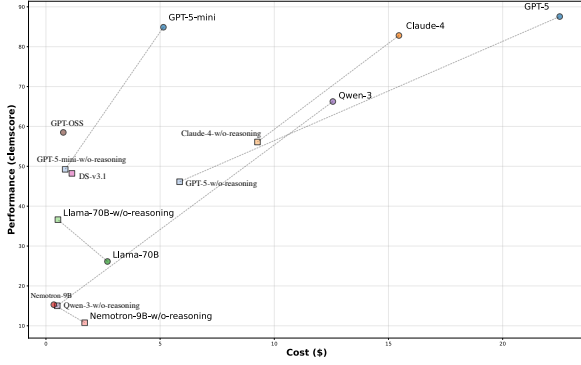


Figure 9: Performance (clemscore) and average number of output tokens for all evaluated models with their reasoning on/off modes for German.

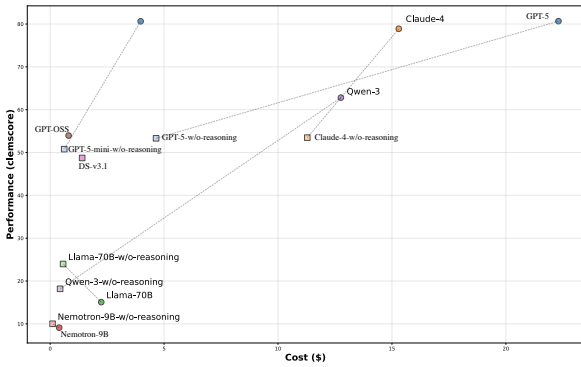


Figure 10: Performance (clemscore) and cost for all evaluated models with their reasoning on/off modes for Italian.

C.2 Experimental Setup

Example grids for the three difficulty levels are given in Figure 14.

C.3 Scoring

The expected distance of two objects on an $w \times h$ grid is calculated as follows:

- Assume two independent variables i and j on one discrete dimension $\{1, 2, \dots, w\}$
- The expected absolute distance between i and j can be calculated as

$$\mathbb{E}[|i - j|] = \frac{1}{w^2} \sum_{i=1}^w \sum_{j=1}^w |i - j|$$

- The double sum evaluates to

$$\mathbb{E}[|i - j|] = \frac{1}{w^2} \cdot \frac{(w-1)(w+1)}{3} = \frac{w^2 - 1}{3w}$$

For two dimensions, we can then calculate the

Euclidean distance of objects o_1 and o_2 as follows:

$$\mathbb{E}[|o_1, o_2|] = \sqrt{\left(\frac{w^2 - 1}{3 \times w}\right)^2 + \left(\frac{h^2 - 1}{3 \times h}\right)^2}$$

For a 7×7 grid, we thus get an expected distance of ≈ 4.19 for two randomly placed objects. To calculate the Expected Distance Sum, we simply multiply it by the number of objects.

This is only an approximation, since we neither take into account the actual grid configuration, nor that two objects cannot have the same location on the same grid.

C.4 Overall Results

C.4.1 Overall Results across levels

As explained in section 3.2, we designed three broad game difficulty levels based on the ratio of empty cells in a grid. We hypothesize that the denser the background grid is, the more likely the target locations clash with occupied cells, and therefore the more likely models get penalized in scoring. In addition, at each difficulty level, there are three sub-levels: either 3, 5, or 7 objects are placed on the grid, on the assumption that the more objects there are, the longer the gameplay traces are, and the less likely that models achieve a high score with mere luck.

Figure 15 plots the main score; clear stratifications of performance across the number of objects are observed in easy and hard levels. When we break down the Main Score into Distance Score and Penalty Score in Figure 16, the stratification at the medium level can also be found.

Games with 3 objects achieve the highest scores, regardless of background empty cell ratios, which corroborates our assumption that games with a low number of objects are easier. However, levels with 5 objects have the lowest Distance Score in the medium level and the lowest Penalty Score in both easy and hard levels, indicating that there are factors other than the number of objects that affect performance.

C.4.2 Overall Results across model properties

Figure 17 illustrates that the Main Scores of commercial models are skewed toward the upper end of the scale, with a clear concentration in the 75100 range, particularly under reasoning mode. In contrast, open models generally score lower, with many results in the 050 range. Notably, the dots are much sparser for open models when reasoning is

TEMPLATE B.1.1

You are given three images, one is called target and the other two are distractors. Your task is to generate a referring expression that best describes the target image while distinguishing it from the two other distractor images. The first image is <IMAGE_POSITION>, the second image is <IMAGE_POSITION>, and the third image is <IMAGE_POSITION>.

Instruction: Describe the target image. Generate the referring expression starting with the tag "Expression: " for the given target image. Omit any other text.

Target image: <IMAGE_PATH>
Second image: <IMAGE_PATH>
Third image: <IMAGE_PATH>

TEMPLATE B.1.2

Expression: \$EXPRESSION\$

TEMPLATE B.1.3

You are given three images. You are also given a referring expression that describes one of the given images. Your task is to select the image that matches the given referring expression. Generate only the number (in text) of the image that matches the given expression by selecting first, second, or third.

TARGET_EXPRESSION

Question: Which image does the expression refer to? Start with the tag "Answer: ", followed by your selection. Omit any other text.

First image: <IMAGE_PATH>
Second image: <IMAGE_PATH>
Third image: <IMAGE_PATH>

TEMPLATE B.1.4

Answer: \$ANSWER\$

(a) Prompt template for Player A (Instruction Giver) in the Reference Game.

(b) Prompt template for Player B (Instruction Follower) in the Reference Game

Figure 11: Reference game prompt templates for both players

on. This is because the Main Score is not recorded when the maximum number of penalties is reached; in these cases, the game will be marked as aborted.

In Figure 18, we further investigate the distribution of Distance Score and Penalty Score. The aborted games, while not shown in the Main Score plot, are represented here with an "x" marker. We set the Penalty Score to 0.5 when the models reach the maximum allowed penalties. If they incur one more penalty, the game is marked as aborted, and the Main Score won't be recorded.

In the Penalty Score subplot, all game plays with Penalty Score < 0.4 are marked with "x" by definition. While the total number of game plays in each of the four lanes is the same, we see that the open reasoning models have the highest share of aborted games, followed by open non-reasoning models. In the Distance Score subplot, we see that the aborted games span widely for open non-reasoning models, indicating some game plays, although achieving a high Distance Score, have pitifully accumulated too many penalties and thus have been marked aborted and taken out of the account of the Main Score. In contrast, commercial models have generally better Penalty Score and Distance Score.

An interesting observation is that, in the Distance

Score subplot, the aborted games tend to reside in lower ranges for non-reasoning models, while the reasoning models might play an aborted game despite achieving a high Distance Score. This is especially prominent for open non-reasoning models, and hints that the reasoning mode, rather than helping in games where these models might achieve a high Distance Score, might introduce noises that lead to more penalties.

C.5 Detailed Analysis

Idea: We hypothesize that Instruction Following ability and Spatial Estimation abilities serve as the foundational pillars of performance, as illustrated in Figure 19. With weaker pillars, model reasoning might not add much to the performance.

Penalties include the Invalid Format Penalty and the Invalid Move Penalty. We plan to use the ratio of Invalid Format Penalty as an agent of Instruction Following ability, sort the models by it, compare the result of the same model with reasoning on or off, and see what percentage the reasoning capability adds to the performance. The assumption is that models with better Instruction Following abilities benefit more from the introduction of the reasoning mode.

TEMPLATE C.1.1

I am your game master, and you are playing a collaborative game with the following grid as game board:

1234567

C	o	o	o				1
o	I	o					2
o	o						3
o	o		o	o			4
							5
o		o	W	T			6
o		o	H				7

* The upper edge displays x-coordinates increasing to the right, and the right edge y-coordinates increasing downward.

* The following objects are randomly placed on your grid: 'W', 'I', 'T', 'C', 'H'.

The other player sees a variation of the game board, where the objects are placed at different random locations. You cannot see the other player's board, and they cannot see yours.

Goal:

Both players need to move the objects on their respective background so that identical objects end up at the same coordinates. You have to communicate with the other player to agree upon a common goal configuration.

Rules:

* In each turn, you can send exactly one of the following two commands:

1. ``SAY: <MESSAGE>``: to send a message (everything up until the next line break) to the other player. I will forward it to your partner.

2. ``MOVE: <OBJECT>, (<X>, <Y>)``: to move an object to a new position, where ``<X>`` is the column and ``<Y>`` is the row. I will inform you if your move was successful or not.

* If you don't stick to the format, or send several commands at once, I have to penalize you.

* If both players accumulate more than 12 penalties, you both lose the game.

* It is vital that you communicate with the other player regarding your goal state! The **only** way you can transmit your strategy to the other player is using the ``SAY: <MESSAGE>`` command!

Moving Objects

* You can only move objects to cells within the bounds of the grid. The target cell must be empty, i.e., it must only contain the symbol 'o'.

* If you try to move an object to a spot that is not empty, or try to move it outside of the grid, I have to penalize you. You get another try.

* Before making a move, double check that the target spot is empty, and does not hold any letter, frame, or line!

End of Game

If you think you reached the goal of aligning all objects, you can ask the other player to finish the game by sending ``SAY: finished?``. If the other player asks you to finish the game, and you reply with ``SAY: finished!``, the game will end.

Both players win if the game ends within 20 rounds, where one round is defined as two players each sending a valid command.

Scoring:

The closer the identical objects are in both game boards, the more points you get. Penalties reduce your points. Can you beat the record?

TEMPLATE C.1.2

Please send a message to the other player to start the game!

TEMPLATE C.1.3

The other player started the game by sending this message:

`"<START_MESSAGE>"`

What is your first command?

(a) Prompt template for Player A in the game Clean Up.

(b) Prompt template for Player B in the game Clean Up.

Figure 12: Clean Up prompt templates for both players

TEMPLATE C.1.4

Penalty: <REASON>

Make sure that your response only contains either SAY: <MESSAGE> or MOVE: <OBJECT>, (<X>, <Y>), and nothing else!

You have collectively accumulated <N> of <M> penalties. Please try again!

<REASON> can be one of the following:

- Your message must not contain anything before the command!
- Your message must not contain anything after the command!
- Your message must not contain anything before or after the command!
- Your message contains more than one command!
- Your message is not in the expected format!
- You must begin the game by sending a message to the other player!

(a) Penalty prompt template for format errors in the game Clean Up

TEMPLATE C.1.5

<REASON>

You have collectively accumulated <N> of <M> penalties. Please try again!

<REASON> can be one of the following:

- Invalid move: (<X>,<Y>) is out of bounds.
- Penalty: (<X>,<Y>) is not empty, but contains '<OBJECT>'.
- Invalid move: Your image has no object with ID '<OBJECT>'.

(b) Prompt templates for invalid moves in the game Clean Up

TEMPLATE C.1.6

<LAST MOVE>

You are currently playing round <R> of maximum <MR>. You have collectively accumulated <N> of <M> penalties. <OTHER PLAYER ACTION>
What is your next command?

<LAST MOVE> can be:

- Your message has been relayed to the other player.
- Moved '<OBJECT>' to (<X>,<Y>) successfully. Your updated grid looks like this:
<GRID>

<OTHER PLAYER ACTION> can be:

- The other player sent this message:
"<MESSAGE>"
- The other player moved an object on their grid.

(c) New turn prompts in the game Clean Up

Figure 13: Intermittent prompts in the game Clean Up

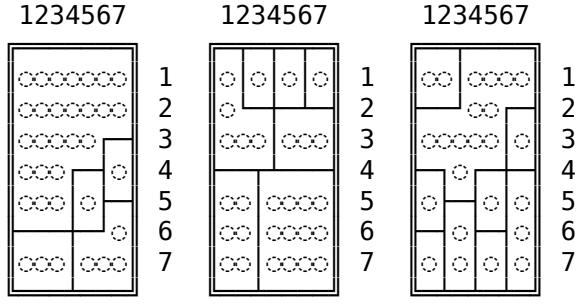


Figure 14: Example grids for *easy*, *medium*, and *hard* levels in Clean Up.

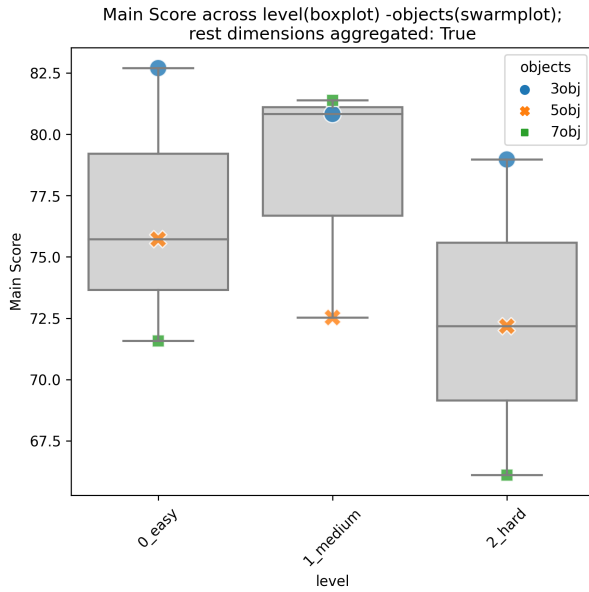


Figure 15: Main Score across levels and sub-levels

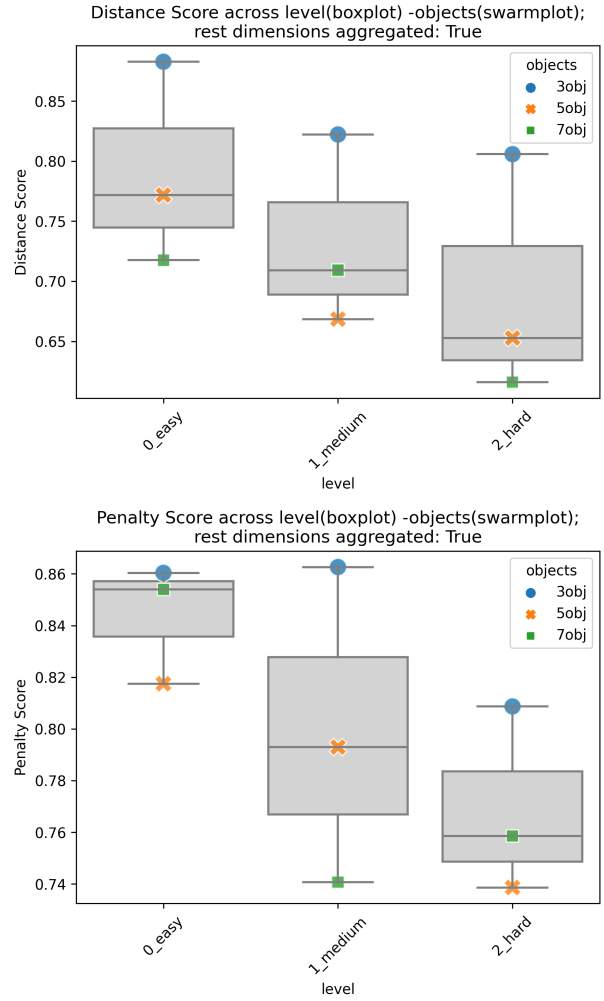


Figure 16: Sub Scores across levels and sub-levels

We plan to apply the same analysis to Invalid Move Penalty.

C.6 Qualitative Samples

Idea: We plan to do qualitative analysis from the following four angles: Spatial Estimations, Strategy, Critical Thinking, Communication & Collaboration, and analyze and compare performance across commercial models (Y/N) and reasoning models (Y/N). For GPT-5, we compare across model size (original / mini) as well.

Here we briefly motivate each of these angles. Concrete examples and analysis will be added later.

C.6.1 Spatial Estimation

From this angle, we check if models can propose valid moving plans. It captures models' Spatial Estimation ability in our grid settings and serves as a foundational pillar of the overall performance. When models are not capable in this regard, they

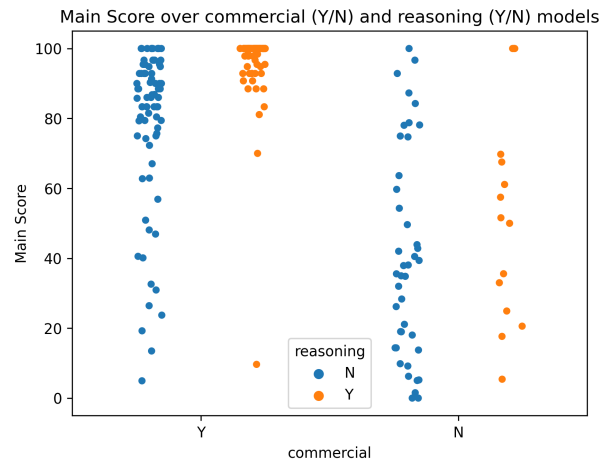


Figure 17: Main Score over different model properties

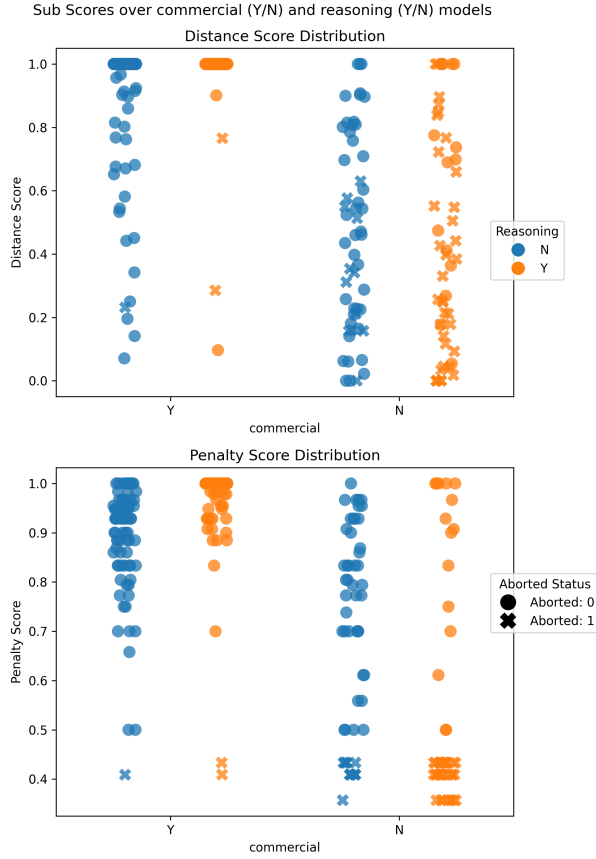


Figure 18: Sub Scores over different model properties

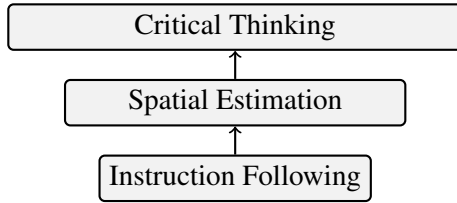


Figure 19: Capability Hierarchical

tend to propose invalid moving plans, and attempts to resolve the wrong movements after getting penalties often conduce to cascades of invalid moving plans, therefore quickly exhaust the number of allowed penalties and end up either aborting the game, or attain a low penalty score.

C.6.2 Strategy

In addition to the crude Spatial Estimation, we check how models come up with moving plans. We have observed some smart strategies and their not-so-smart counterparts. In some game plays, one model uses the initial coordinates of the objects to instruct the other model as target locations, thus avoiding penalties of moving objects to cells occupied by the background grids to a large extent; while in some other game plays, models appear to

exchange initial coordinates just for the sake of exchanging, without making use of this information. Another example is in some game plays, models choose the most sparse row as the target, and lay objects one by one on it; while in some other game plays, the models unnecessarily chose geometric patterns to lay the objects, such as on the diagonal, disregarding the occupied cells, and run into complexities that they could have dodged.

C.6.3 Critical Thinking

Some models show the ability to critically reason about the moving plan that either itself or the other player proposed. They either refute the plan immediately when it's proposed, or take a sharp brake when they are just about to execute the invalid moving plan. This is, as expected, more frequent in reasoning models compared to non-reasoning models. However, on some occasions, we see pseudo-reasoning: a model rejects a correct plan, utters reasoning-like sentences, and proposes another plan that sometimes is correct, sometimes is wrong.

C.6.4 Communication & Collaboration

The above three dimensions positively correlate with the model performance: The better the models are at them, the higher the scores. This dimension, however, is different. Specifically, we observe how models communicate with each other to recover from invalid moves due to target locations being occupied. Some models go at lengths to execute their alternative plans without letting their partner know, they might still have a good score, simply because both player 1 and player 2 adopt the exact same strategy, for example, move the object to one spot next to the original target location. This leads to inflated scores that can be debunked when two models of different default recovery strategies, or one model and one human, play against each other.

D Air Balloon Survival - Game Details

D.1 Prompt Templates

The prompt template for both players of the Hot Air Balloon Survival game is given in Figure 20. Parse error prompts handed to players when their responses did not follow the format instructions are given in Figure 21, and game error prompts for invalid actions are given in Figure 22.

Substrings Figure 20 marked by \$. . . \$ are placeholders which get replaced depending on the game

TEMPLATE D.1.1

You are participating in a collaborative negotiation game.

Together with another participant, you must agree on a single set of items that will be kept. Each of you has your own view of how much each item matters to you (importance). You do not know how the other participant values the items. Additionally, you are given the effort each item demands. You may only agree on a set if the total effort of the selected items does not exceed a shared limit:

LIMIT = \$LIMIT\$

Here are the individual item effort values:

Item effort = \$ITEM_WEIGHTS\$

Here is your personal view on the importance of each item:

Item importance values = \$UTILITY_SCALE_PLAYER\$

Goal:

Your goal is to negotiate a shared set of items that benefits you as much as possible (i.e., maximizes total importance to YOU), while staying within the LIMIT. You are not required to make a PROPOSAL in every message - you can simply negotiate as well. All tactics are allowed!

Interaction Protocol:

You may only use the following structured formats in a message:

PROPOSAL: {'A', 'B', 'C', }
Propose keeping exactly those items.

REFUSE: {'A', 'B', 'C', }
Explicitly reject opponent's proposal.

ARGUMENT: {'...'}
Defend your last proposal or argue against the player's proposal.

AGREE: {'A', 'B', 'C', }
Accept the opponent's proposal which ends the game.

\$STRATEGIC_REASONING_FORMAT\$

Rules:

You may only AGREE on a proposal the other party has logged via PROPOSAL.

You may only REFUSE a proposal the other party has logged via PROPOSAL.

Total effort of any PROPOSAL or AGREE set must be $\leq$ LIMIT.

Do NOT reveal your hidden importance scores.

A tag in a structured format must be followed by colon and whitespace. The argument must be a python set containing 0 or more strings.

So, it must be of the form TAG: {...}

Strictly follow the interaction protocol and DO NOT write anything beyond the given structure.

The game ends when one side gives an AGREE to a PROPOSAL made by the other player.

The content in your response which can be handed to the other player has to be non-empty.

Only proposals which have been logged via the PROPOSAL format structure and which haven't been refused via REFUSE are active.

\$REQUIRE_ARGUMENT\$

\$STRATEGIC_REASONING_RULE\$

Figure 20: Initial prompt template of Hot Air Balloon. Note that for the second player we append newline and the phrase 'You will now receive the first message of the other player.' at the very end of the initial prompt.

instance. Note that the last three placeholders may also be replaced by the empty string depending on the game settings. Here we list the replacements for placeholders in the initial prompt:

- `$LIMIT$` maximum weight for a deal.
- `$ITEM_WEIGHTS$` mapping from items to weights.
- `$UTILITY_SCALE_PLAYER$` mapping from items to preference values for a given player.
- `$STRATEGIC_REASONING_FORMAT$`

STRATEGIC REASONING: { '...'}
Describe your strategic reasoning or anticipation explaining your choice of action. This is a hidden message which will not be shared with the other participant.
- `$REQUIRE_ARGUMENT$` You must include the ARGUMENT format at least once somewhere in all of your messages.
- `$STRATEGIC_REASONING_RULE$`

You must include the STRATEGIC REASONING format only once at the very beginning of every one of your messages and not more often. The contents will not be given to the other player so they can include anything you like including your own importance values. Here you should reason step by step to come up with your next move.

D.2 Experimental Setup

D.3 Overall Results

D.4 Detailed Analysis

D.5 Qualitative Samples

TEMPLATE D.1.2

Your response did not start with the proper strategic reasoning tag at the very beginning of your response.
The very first structured format must be of the form STRATEGIC REASONING: {...}. Try again.

(a) Parse error prompt in case strategic reasoning tag at the beginning of a response is missing. Only applies when strategic reasoning tag is required

TEMPLATE D.1.3

Your response did not contain an argument.
You must include the structured format ARGUMENT: {...} somewhere in your response. Try again.

(b) Parse error prompt in case an argument is missing. Only applies when argument tag is required

TEMPLATE D.1.4

Your response contained an untagged sequence or you used STRATEGIC REASONING more than once.
You may only use the structured formats as explained in the initial message.
They must all be of the form TAG: {...}.

(c) Parse error prompt in case the response contains an untagged sequence.

TEMPLATE D.1.5

Your response only contained a strategic reasoning tag.
You must at least include one more valid tag in your response, so that the other player receives a message. Try again.

(d) Parse error prompt in case response only contained a strategic reasoning tag. Only applies when argument tag is not required.

TEMPLATE D.1.6

You used a PROPOSAL tag, but did not provide a valid python set containing strings as arguments, e.g. {'A', 'B', 'C', ...}. Try again.

(e) Parse error prompt in case of an invalid python set as argument.

Figure 21: Parse error prompts handed to player when a response did not follow the instructions on structured formats.

TEMPLATE D.1.7

You refused a proposal which is not active.
Proposals are only active if they have been
logged by the other player via PROPOSAL and
have not been deactivated by you via REFUSE.
Try again.

(a) Game error prompt in case a non-active was refused.

TEMPLATE D.1.8

You made more than one agreement.
Final deals cannot be ambiguous. Try again.

(b) Game error prompt in case agreement to a deal was ambiguous.

TEMPLATE D.1.9

You agreed to a proposal which is not active.
Proposals are only active if they have been
logged by the other player via PROPOSAL and
have not been deactivated by you via REFUSE.
Try again.

(c) Game error prompt in case .

TEMPLATE D.1.10

Your proposal includes items which are not in
the game. Try again.

(d) Game error prompt in case a non-active proposal was agreed to.

Figure 22: Game error prompts handed to player when a response did not follow the instructions on the rules of the game.